Beijing Lin

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EDUCATION

University of British Columbia(Permanent resident) 09/2024-Present

Bachelor of Electrical Engineering(Co-op)

Relevant courses: Circuit Analysis I, Data Structures and Algorithms, Computing Systems I

TECHNICAL EXPERIENCE

Real-Time Tron Light Cycle Game 11/2025-12/2025

- Developed a real-time two-player game in C running on an FPGA platform using a memory-mapped VGA graphics interface
- Implemented hardware timer interrupts and key interrupts to handle game timing, player input, and deterministic movement updates
- Built collision detection by reading VGA frame buffer pixel data, enabling real-time obstacle interaction
- Debugged timing and memory errors through visual debugging, boundary validation, and hardware state indicators

FPGA Combination Lock System 01/2025-02/2025

- Designed and built an autonomous object-retrieval system for a multi-round competition
- Programmed and tested Arduino-based servo-controlled grippers using sensors for autonomous grasping and transport

STUDENT DESIGN TEAMS

UBC Rocket, Avionic Hardware & System Validation 09/2025-Present

Avionic Hardware:

- Designed and validated the rocket's PYRO board using Altium, analyzing component datasheets and simulating circuits with LTspice
- Developed and tested embedded functionality with STM32 microcontroller, conducting hands-on system verification

System Validation:

- Brought up peripheral configuration and pin mapping in CubeMX (SPI, GPIO, interrupts), and validated the interface through unit-level firmware tests and debug instrumentation (UART, logic-level checks)

UBC Open Robotics, Automatic Guitar Tuner Member 01/2026-Present

- Configured signal conditioning and sampling parameters to capture guitar frequencies
- Designed a custom PCB for microphone and ESP32 signal routing within mechanical enclosure size constraints
- Integrated power architecture using LiPo battery and DC-DC conversion modules for portable operation

TECHNICAL SKILLS

Programming & HDL: C, Java, Python, FPGA Verilog/SystemVerilog, RISC-V Assembly

Software & Tools: MATLAB, Quartus, Questa, Visual Studio, Windows Powershell, MS Office(Excel), Fusion 360, SolidWorks

Electrical & Circuits: Altium Desiger, PCB Assembly, LTspice, STM32 MCU, SMD Soldering, Oscilliscope, Multimeter, Signal Generator, Circuit Design & Analysis

AWARDS

Award 1:AP Scholar with Distinction Award 07/2024

Award 2: The Governor General's Academic Medal 2023-2024